DEEP LEARNING LABORATORY

EXPERIMENT 10

L2, DROPOUT AND EARLY STOPPING COMPARISON

Aim

To implement and compare L2 regularization, Dropout, and Early Stopping using a neural network, and evaluate their effects on validation loss and validation accuracy.

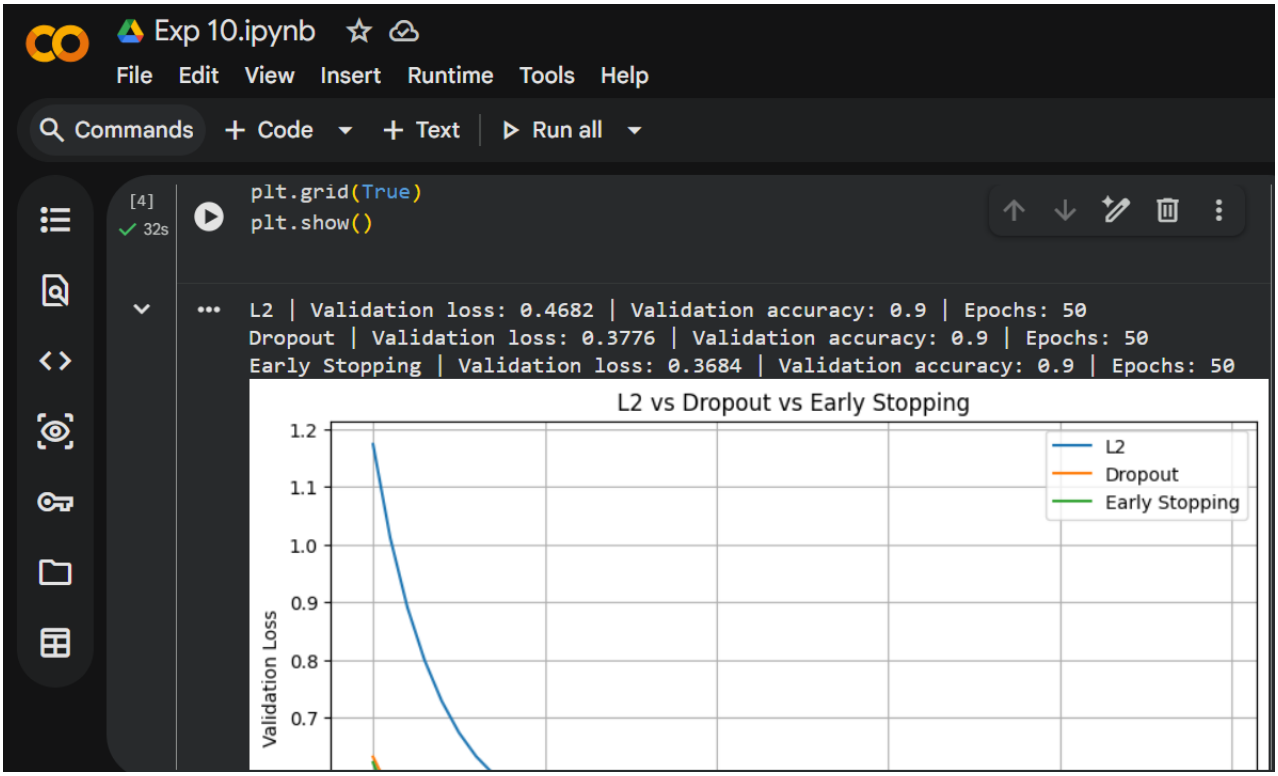
Output

The Colab output reports the following validation metrics: L2 — validation loss 0.4682 and accuracy 0.9; Dropout — validation loss 0.3776 and accuracy 0.9; Early Stopping — validation loss 0.3684 and accuracy 0.9. Each method ran for 50 epochs.

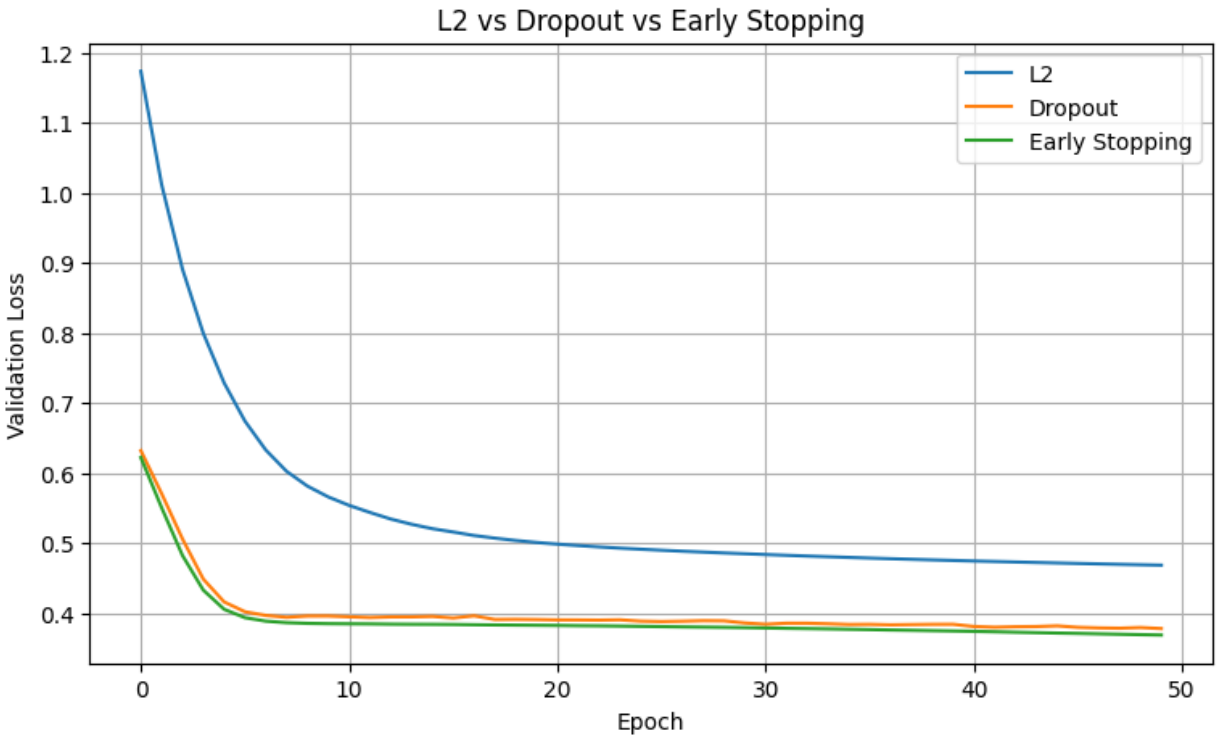
Observed results

Method	Validation loss	Validation accuracy	Epochs
L2	0.4682	0.9000	50
Dropout	0.3776	0.9000	50
Early Stopping	0.3684	0.9000	50

Colab output screenshot



Validation-loss comparison graph


L2 vs Dropout vs Early Stopping
L2
Dropout
Early Stopping
Validation Loss
Epoch
1.2
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0
10
20
30
40
50